

Yash Prakash Mishra

Senior Software Engineer — Backend (Node.js · TypeScript · PostgreSQL)

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PROFESSIONAL SUMMARY

Backend engineer with 5+ years building high-volume, money-moving systems in **Node.js** and **TypeScript** on **PostgreSQL**, for SaaS products running 24/7 on AWS. Designed and owned a **gateway-agnostic payments platform** across four providers (Stripe, Razorpay, Affirm, Jodo) with **idempotent transaction handling**, webhook reconciliation and circuit-breaker failover, and a **loan management system serving 10,000+ active loans** for OLX Autos in Chile. Strong on **OOP** and **design patterns**, **queue-based architecture** (Kafka, BullMQ, SQS), JWT auth and input validation, and **multi-tenant architecture** taken from B2C to 5+ enterprise clients. Fully remote for 5 years, collaborating asynchronously with product, design, QA and frontend across India, LatAm and the Middle East.

CORE SKILLS

Languages & Runtime	TypeScript, JavaScript (ES6+), Node.js, SQL, C++
Backend & APIs	Express.js, NestJS, Fastify, RESTful API design, input validation & schema guards, OpenAPI/Swagger, BFF pattern
Databases	PostgreSQL (query optimization, data modeling, zero-downtime migrations), MySQL, ClickHouse, MongoDB, Redis, Sequelize, Prisma
Queues & Events	Apache Kafka, BullMQ, AWS SQS, event-driven architecture, webhooks, idempotency & event deduplication, Server-Sent Events, Socket.io
Architecture & Scale	Microservices, distributed systems, multi-tenant architecture, idempotent API design, saga pattern, circuit breaker, Strategy / Factory / Singleton / Observer, HLD & LLD, high transaction volumes, concurrency control, graceful degradation
Auth & Security	JWT issuance, rotation & validation, Passport.js, OAuth2, centralised RBAC, audit logging
Cloud & DevOps	AWS (EC2, S3, SQS, KMS, IAM, CloudFront), Docker, Kubernetes (working knowledge), CI/CD (GitHub Actions, Jenkins), Apache Airflow
Observability & Testing	ELK Stack, structured logging, distributed tracing, Jest, Supertest, Mocha, integration testing, Postman
Frontend (secondary)	React.js, Next.js, Redux

PROFESSIONAL EXPERIENCE

Novatr — Senior Software Engineer (Backend Developer → Senior Software Engineer) Aug 2023 – Present
Remote · Y Combinator-backed EdTech SaaS, professional upskilling

Stack: Node.js, TypeScript, NestJS, Express.js, PostgreSQL, ClickHouse, Redis, Apache Kafka, BullMQ, AWS (EC2, S3, SQS, KMS, IAM), Docker

Payments & money movement

- Architected a **gateway-agnostic payments platform** in Node.js + TypeScript integrating four providers (Razorpay, Stripe, Affirm, Jodo), using **Strategy and Factory patterns** for runtime gateway selection so new gateways onboard without touching core logic; owned the integration POC for each provider.
- Built **idempotent transaction handling** across every payment path, making retries, duplicate webhooks and client re-submissions safe under concurrent load.
- Designed and owned a **centralised webhook handler** for all four gateways with self-healing retry logic, **event deduplication** and status reconciliation over Apache Kafka — cutting manual error resolution by **80%**.
- Applied **circuit breaker** (opossum) and retry strategies (axios-retry) for intelligent failover between gateways, keeping checkout available through provider outages.
- Delivered a **referral and rewards payout platform** (Express.js, PostgreSQL, Kafka) with PAN verification, fraud prevention, audit trails and Xoxoday programmatic payouts — **+25% referral conversions** and zero manual reward processing.

Scalability, data & reliability

- Introduced **Kafka-based event pipelines** (with BullMQ for in-process jobs) for payments, user lifecycle events and transactional email, decoupling services and raising fault tolerance platform-wide.
- Owned a **zero-downtime data migration** of payments history from a legacy system onto a ClickHouse-backed unified platform via Node.js ETL, with schema validation, integrity checks and a tested rollback path.
- Built a real-time quiz engine (Socket.io, Redis) tuned for **low-latency concurrent interactions** across live cohort sessions, with graceful degradation under load.

Platform & API engineering — Novatr

- Led the **B2C-to-B2B transition** by architecting a **multi-tenant** system (NestJS, PostgreSQL) with tenant-level data isolation, per-tenant configuration and enterprise onboarding workflows — **onboarded 5+ enterprise clients** as technical point of contact.
- Architected a **centralised RBAC service** (Node.js, TypeScript, PostgreSQL) across 5+ internal tools with role inheritance, permission scoping and audit logging — **60% fewer access incidents**.
- Built a multi-layer **JWT authentication service** (Passport.js) handling token issuance, rotation and validation across Admin, LMS and BFF surfaces.
- Designed a CMS backend (NestJS, PostgreSQL) with content lifecycle workflows and versioning — **85% faster cohort creation**, recurring on-call incidents eliminated.
- **Led a team of 5 engineers** on architecture decisions, code reviews and delivery timelines; partnered with product, design, QA and frontend from discovery through deployment, and was technical point of contact for enterprise clients translating business requirements into architecture.

OLX Autos — Software Engineer

May 2022 – May 2023

Remote · Used-car financing infrastructure for Latin America (FinTech)

Stack: Node.js, TypeScript, Express.js, PostgreSQL, MySQL, Sequelize, REST APIs, Docker, AWS

- Built and owned foreclosure and manual repayment flows in the **Loan Management System** (Node.js, Express.js, PostgreSQL), automating loan closure and repayment tracking for **10,000+ active loans in Chile** and replacing a fully manual process.
- Designed **loan lifecycle state machine APIs** with **input validation guards at every transition**, cutting manual intervention by **40%** and making state changes auditable.
- Designed and owned risk analysis and credit scoring workflows in the **Loan Origination System**, reducing credit approval time by **30%** and improving decision auditability.
- Built a reusable PDF generation library (PDFKit, Handlebars) for programmatic invoices inside loan disbursement flows — fewer invoice errors, 10+ hours saved weekly; served as technical point of contact for the Chile operations team **asynchronously across a 9+ hour time-zone gap**.

Lenskart — Software Engineer (Frontend)

Apr 2021 – Apr 2022

India's largest eyewear retailer · Exit: project concluded on successful UAE market launch

Stack: React.js, Redux, TypeScript, Next.js, Node.js (BFF), Webpack, REST APIs

- Led the rebuild of core commerce flows (homepage, cart, checkout, order tracking) in React.js, Redux and TypeScript for the UAE launch — **30% faster launch** with improved Lighthouse scores, as primary frontend point of contact for go-live.
- Consumed and shaped **Node.js BFF APIs** for the new storefront, defining response contracts with backend teams — the work that moved me fully into backend engineering; also cut initial JS payload via code splitting and bundle analysis (Webpack) and integrated reCAPTCHA to reduce fraudulent checkouts.

SELECTED PROJECT

BullMQ Dashboard — open-source queue operations tool

2025

Next.js, TypeScript, Node.js, Redis, ioredis, BullMQ, Express.js, Docker, Server-Sent Events · github.com/xcalibre1/bullmq-dashboard

- **Problem:** existing tools give visibility into **queue-based systems** but no operational control — no retrigger with an edited payload, no bulk retry, no live updates.
- Built a Next.js + TypeScript dashboard with **Server-Sent Events** for live queue stats, an editable JSON payload editor and a 24-hour throughput chart in pure SVG; App Router API routes backed by a **Redis singleton** (ioredis) and a cached BullMQ Queue instance for near-zero per-request overhead, with Docker Compose for one-command self-hosting. Adopted by engineering teams globally.

EDUCATION

B.Tech, Computer Science — Rajkiya Engineering College, Sonbhadra

Jul 2016 – Jun 2020

ADDITIONAL

- **Open source:** author and maintainer of BullMQ Dashboard (Node.js, TypeScript), used by engineering teams globally for queue operations.
- **Remote-first:** 5 years fully remote across distributed teams in India, Latin America and the Middle East, owning work asynchronously and communicating in writing. **Interests:** distributed systems, payments infrastructure, FinTech, developer tooling.